

Acknowledgements

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Name: _____

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Line Following Robot

Objective:

The aim of this project is to design and develop a **smart line following and robot** using **Arduino, IR sensors**. The primary objective is to create a robotic system capable of **autonomously navigating and following a line** while intelligently detecting along its route. The **IR sensors** are used to sense and follow the black or white line on the surface, processing input signals from sensors and controlling the motors accordingly. This project demonstrates the integration of sensors and control logic to achieve **autonomous navigation**, which can be applied in industrial automation, delivery systems, and smart mobility solutions.

Components Required and Cost:

Name of Components	Quantity	Component's Work	Price
Arduino Uno with Cable	1	Microcontroller to control the robot	460.00
IR Sensor Module	2	For detecting the black/white line	100.00
L298N Motor Driver	1	To control the DC motors	190.00
Switch	1	Power on or off	20.00

Battery Pack (12V)	3	Power supply for the robot	150.00
Chasis	1	To build base of the robot	200.00
Wheels with motor	3	To Drive the Robot	450.00
Jumper Wires	9-10	For making connections	50
TOTAL =			1620.00

Working:

The working of the **Smart Line Following** can be divided into three main modes — **Line Following Mode**, and **Continuous Operation Mode**.

1. Line Following Mode:

In this mode, the **IR (Infrared) sensors** play a major role. The robot is designed to follow a black line drawn on a white surface. The IR sensors continuously sense the surface beneath the robot. When the **left sensor detects the black line**, the robot understands that it has deviated from the path, so it turns **left** to correct its direction. Similarly, if the **right sensor detects the black line**, the robot turns **right**. When **both sensors detect the black line simultaneously**, the robot moves **straight forward** along the path. This process allows the robot to stay aligned with the predefined route.

2. Continuous Operation Mode:

The robot continuously operates by combining both line following features. It keeps following the line smoothly in its path. The robot automatically resumes line following. The Arduino microcontroller coordinates all sensor data and motor actions, ensuring smooth direction changes and speed control. This makes the robot highly efficient, autonomous, and suitable for applications like **automated transport, industrial delivery systems, and smart navigation systems**.

Procedure: (Assembling the line following Robot):

- Gather components:
 - Collect all the component listed above, ensuring they are functionable.
- Prepare robot chassis & assemble the components. □ Fix Bo motors on the chassis (rear side).
 - Attach wheels to the motor securely.
 - Mount a free rotating caster wheel in the front.
 - Place the battery on the chassis using a holder or zip ties.
- Connect motors to L298N motor driver:
 - Left motor wires – C298N OUT1 & OUT2
 - Right motor wires – C298N OUT3 & OUT4 4) L298N
 - 12V battery +ve – L298N “12V” pin
 - Battery -ve (4ND) – L298N “4ND” 5) L298N Enable & control pins (to Arduino):
 - END (speed control) – Audino Pin D3
 - ENB (speed control) – Audino Pin D6
 - IN1, IN2, IN3, IN4 – Audino Pin D4, D5, D7, D8
- Connect IR sensor:

- Left IR sensor out – Arduino D2
- Right IR sensor out – Arduino D3
- VCC and GND – Arduino 5V & GND

5. Upload code to Arduino:

- Connect Arduino to PC via USB.
- Upload the combined code for line following and obstacle avoiding. □ Test the motor movement and sensor detection.

6. Final testing and adjustment:

- Check if motors rotate correctly (adjust motor wires if needed).
- Place the robot on black and white track & test line following.
- Place an obstacle in front of the robot to check if the robot stops or not and whether the sensor is working.

Potential Future Enhancements for Line Following Robot:

1. Advanced Obstacle Avoidance:

- Implement multiple ultrasonic sensors for 360° obstacle detection.
- Integrate LiDAR or camera-based vision for better obstacle recognition.

2. Wireless Control & Monitoring:

- Add Bluetooth or Wi-Fi modules for remote control via a mobile app.
- Add speed regulator for better control of the robot.

3. Multi-Terrain Adaptability:

- Use better suspension and all-terrain wheels for uneven surfaces.
- Develop an adaptive control algorithm to handle different track materials like used in modern cars by the name of terrain modes.

Arduino Code:

```
#define IR_SENSOR_RIGHT 11
#define IR_SENSOR_LEFT 12
#define MOTOR_SPEED 180

// Right Motor
int enableRightMotor = 6;
int rightMotorPin1 = 7;
int rightMotorPin2 = 8;

// Left Motor
int enableLeftMotor = 5;
int leftMotorPin1 = 9;
int leftMotorPin2 = 10;

void setup()
{
  // Change PWM frequency for smoother motor control
  TCCR0B = TCCR0B & B11111000 | B00000010;

  pinMode(enableRightMotor, OUTPUT);
```

```

pinMode(rightMotorPin1, OUTPUT);
pinMode(rightMotorPin2, OUTPUT);

pinMode(enableLeftMotor, OUTPUT);
pinMode(leftMotorPin1, OUTPUT);
pinMode(leftMotorPin2, OUTPUT);

pinMode(IR_SENSOR_RIGHT, INPUT);
pinMode(IR_SENSOR_LEFT, INPUT);

rotateMotor(0,0);
}

void loop()
{
  int rightIRSensorValue = digitalRead(IR_SENSOR_RIGHT);
  int leftIRSensorValue = digitalRead(IR_SENSOR_LEFT);

  // Both sensors detect BLACK line → move straight
  if (rightIRSensorValue == LOW && leftIRSensorValue == LOW)
  {
    rotateMotor(MOTOR_SPEED, MOTOR_SPEED);
  }

  // Right sensor on WHITE → turn right
  else if (rightIRSensorValue == HIGH && leftIRSensorValue == LOW)
  {
    rotateMotor(-MOTOR_SPEED, MOTOR_SPEED);
  }

  // Left sensor on WHITE → turn left
  else if (rightIRSensorValue == LOW && leftIRSensorValue == HIGH)
  {
    rotateMotor(MOTOR_SPEED, -MOTOR_SPEED);
  }

  // Both sensors on WHITE → stop
  else
  {
    rotateMotor(0, 0);
  }
}

void rotateMotor(int rightMotorSpeed, int leftMotorSpeed)
{
  // Right Motor Direction
  if (rightMotorSpeed < 0)
  {

```

```

    digitalWrite(rightMotorPin1, LOW);
    digitalWrite(rightMotorPin2, HIGH);
}
else if (rightMotorSpeed > 0)
{
    digitalWrite(rightMotorPin1, HIGH);
    digitalWrite(rightMotorPin2, LOW);
}
else
{
    digitalWrite(rightMotorPin1, LOW);
    digitalWrite(rightMotorPin2, LOW);
}

// Left Motor Direction
if (leftMotorSpeed < 0)
{
    digitalWrite(leftMotorPin1, HIGH);
    digitalWrite(leftMotorPin2, LOW);
}
else if (leftMotorSpeed > 0)
{
    digitalWrite(leftMotorPin1, LOW);
    digitalWrite(leftMotorPin2, HIGH);
}
else
{
    digitalWrite(leftMotorPin1, LOW);
    digitalWrite(leftMotorPin2, LOW);
}

analogWrite(enableRightMotor, abs(rightMotorSpeed));
analogWrite(enableLeftMotor, abs(leftMotorSpeed));
}

```

Conclusion:

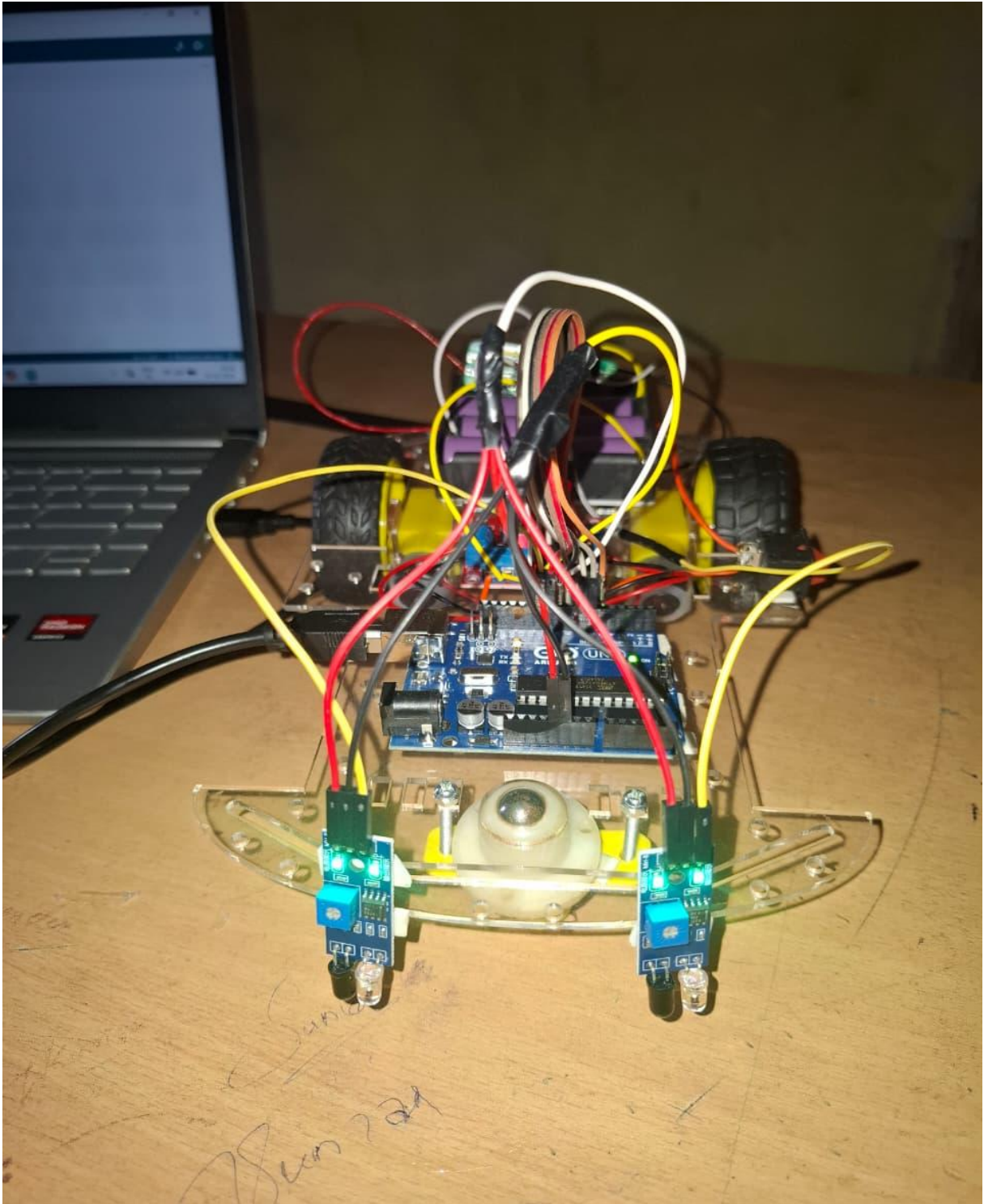
This project successfully demonstrates the working of an autonomous mobile robot that can follow a predefined path using IR sensors. Through this project we learned: • How to use an Arduino uno for controlling outputs. • The basic working of IR sensor, L298N motor driver. • The logic behind following predefined path. This project provides hands on experience on both hardware assembly and microcontroller programming.

Through this project we learned:

1. How to use an Arduino uno for controlling outputs.
2. The basic working of IR sensor, L298N motor driver.

3. The logic behind following predefined.

This project provides hands on experience on both hardware assembly and microcontroller programming.



References:

❖ **Books**

Basic Robotics

Exploring Arduino

❖ **Website**

<https://www.robu.in>

<https://projecthub.arduino.cc>